A LEVEL FUNCTIONS QUESTION

Question: Programming Using Functions – Processing Examination Marks

A school requires a computer program to process examination marks for a class. The program must allow the user to enter the number of students and their examination marks. The marks must be stored in an array.

Write a modular program that uses user-defined functions to process the marks entered. Your program should perform the following tasks:

(a) Prompt the user to enter the number of students and their examination marks, then store the marks in an array. (4 marks)

(b) Write a procedure to display all the marks entered. (2 marks)

(c) Write a function called **FindHighestMark** that accepts the marks array and returns the highest mark. (3 marks)

(d) Write a function called **FindLowestMark** that accepts the marks array and returns the lowest mark. (3 marks)

(e) Write a function called **CalculateTotal** that accepts the marks array and returns the sum of all the marks entered. (3 marks)

(f) Write a function called **CalculateAverage** that accepts the total marks and the number of students, then returns the average mark. (3 marks)

(g) Write a function called **CountMarks** that returns the number of marks stored in the array. (2 marks)

(h) Call the procedure and functions from the main program and display all the results with appropriate output messages. (3 marks)

NB: Programming requirements:

1. Use meaningful and descriptive variable names.
2. Make appropriate use of parameters and return values when defining functions.
3. Avoid repeating code unnecessarily.
4. Display appropriate messages so that the user can understand each result.
5. Use an array to store the marks.

Total: 23 marks